

SAGAR MISHRA

Full-Stack Engineer | React, Next.js, Node.js, TypeScript, Django, PostgreSQL

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Full working rights in Australia.

Full-stack engineer with two years shipping production code at Programiz Pro, a coding-education platform with over 100,000 paid learners. I built the course flow and learner dashboard every paid learner works through, and the engine that issues their completion certificates. Most recently I led four engineers at Jobs.ai, where I owned authentication and the employer workspace and built the marketplace search both sides query. On my own time I have designed and shipped two systems from data model to production: a warehouse platform with lot-level inventory and state-machine fulfilment, and a multi-tenant storefront that runs on top of it. In both roles I led: a team of six on Programiz Reports, four engineers at Jobs.ai.

EXPERIENCE

Software Engineering Intern (Team Lead), Jobs.ai

Job and task distribution platform.

Melbourne, Australia | Feb 2026 – May 2026

- Led four engineers as team lead. I set the technical direction, ran code review, and nothing merged until I had read it.
- Owned authentication and the role-based access model that separates employers from job seekers, which every permission check in the rest of the platform reads from.
- Designed the employer workspace end to end, so an employer posts a job or task and then moves applicants through review, shortlist and rejection without leaving it.
- Built the marketplace search on a single index that serves both sides, so employers looking for people and seekers looking for work query the same system from opposite directions.
- Chose cursor-based pagination over offset for the job and task feeds. Read cost stays flat as the dataset grows rather than degrading with page depth, and the feed stays scoped to active dates.
- Designed the allocation logic that spreads available workers across open jobs and tasks by demand, so no job sits unstaffed while another is oversubscribed.

Software Engineer, ParewaLabs (Programiz)

Coding-education platform, programiz.com and programiz.pro. Over 100,000 paid learners.

Kathmandu, Nepal | Oct 2021 – Nov 2023

- Led Programiz Reports with a team of six on a Scrum cycle. I gathered the requirements, owned the scope, ran sprint planning, and carried it from proposal to release.
- Built the course flow and learner dashboard on programiz.pro, the surface all 100,000+ paid learners move through to finish a course.
- Built the certificate engine that issues a verified completion certificate to every learner who finishes a course.
- Worked on the in-browser Python and JavaScript compilers on programiz.com, where visitors run code without installing anything.
- Integrated sensAI, the in-product tutor, so learners get help at the point in an exercise where they are stuck rather than after they give up.
- Instrumented landing pages with Mixpanel and Microsoft Clarity, then rewrote the pages against what the behaviour data showed.

PROJECTS

Nexus, Warehouse Management Platform

Personal project. Built after running an anime-apparel side business where stock went missing and fulfilment was manual. React, Next.js, Node.js, PostgreSQL, Redis.

Feb 2025 – Jan 2026

- Modelled the warehouse as a labelled tree. Any item resolves to a location in seconds, which is the missing-stock problem the project was built to end.
- Modelled product drops as a state machine covering supplier onboarding, lot negotiation, intake and lot-level pricing. Nothing advances out of order.
- Built lot-aware replenishment with a return-to-order path, so a refund or exchange resolves back to the correct inventory state instead of drifting.
- Wrote the fulfilment pipeline that batches retail orders on a timer and tracks each one to completion as its own state machine.

Aroma, Multi-tenant Clothing E-commerce

Personal project. A storefront layer that sits on Nexus, so sellers run their own shop on shared infrastructure. React, Next.js, Node.js, PostgreSQL.

Mar 2024 – Jan 2026

- Designed the tenancy model so each seller runs an isolated storefront on one shared backend, drawing the line between what is shared and what a tenant controls.
- Built the buyer path front to back: search and product pages through cart, reservation, checkout and order history.
- Split rendering by page: server-side rendering and static generation for the catalogue pages that have to be indexed and paint fast, client fetching for anything that changes per user, with React Query holding the cache and Zustand holding cart state.
- Measured the result in Lighthouse rather than assuming it. Every page renders in under two seconds.
- Added a post-purchase order-status view. Sellers who trade only through social media cannot offer that visibility, and the trust gap costs them sales.

EDUCATION

Victorian Institute of Technology

Master of Information Technology and Systems, Software Engineering

Melbourne, Australia

Feb 2024 – Jul 2026

Softwarica College of IT and E-commerce

BSc (Hons) Software Engineering

Kathmandu, Nepal

Nov 2018 – Oct 2021

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Java, C#, Kotlin, SQL, HTML, CSS

Frameworks & Libraries: React, Next.js, Node.js, Express, Django, React Query, Zustand, Drizzle, Mongoose, Sequelize

Data: PostgreSQL, MySQL, MongoDB, Redis, relational schema design, data modelling

Architecture: REST APIs, multi-tenancy, state machines, microservices, monoliths, sharding, server-side rendering, static generation, role-based access control, caching

Cloud & DevOps: AWS (EC2, RDS, S3, Lambda), Docker, containerisation

Practices: Agile delivery (Scrum), sprint planning, code review, requirements gathering, product analytics and instrumentation

Tools: Git, Jira, Asana, Directus, Mixpanel, Microsoft Clarity

AI-ASSISTED ENGINEERING

I use agentic coding tools to delegate work and then check it. A light model to explore, a stronger one to plan and design, a mid one to implement. I read and test every change before it goes in, and if I do not understand a piece of code it does not merge. The tools speed up boilerplate and exploration. The design calls and the quality stay mine.

REFERENCES

Available on request. My referees are a principal engineer and the founder at Programiz, both of whom I reported into directly.